HUJI · 769001 · DYNAMICS AND CODING IN THE BRAIN

Week 01

The Brain, the Rate Model, and Linear Dynamics

neurons · spikes · synapses · rate model · scales · $\dot{n} = an$ · convolution · eigenvectors

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PART A · LECTURE 1

Introduction: Neurons, Scales and Questions

SECTION 01

Neurons

The brain is made of cells, like every other organ. The cells that process information are called neurons. There are about 10^{11} of them in a human brain [LMLF 1.1.1].

> 1.1 The three parts of a neuron

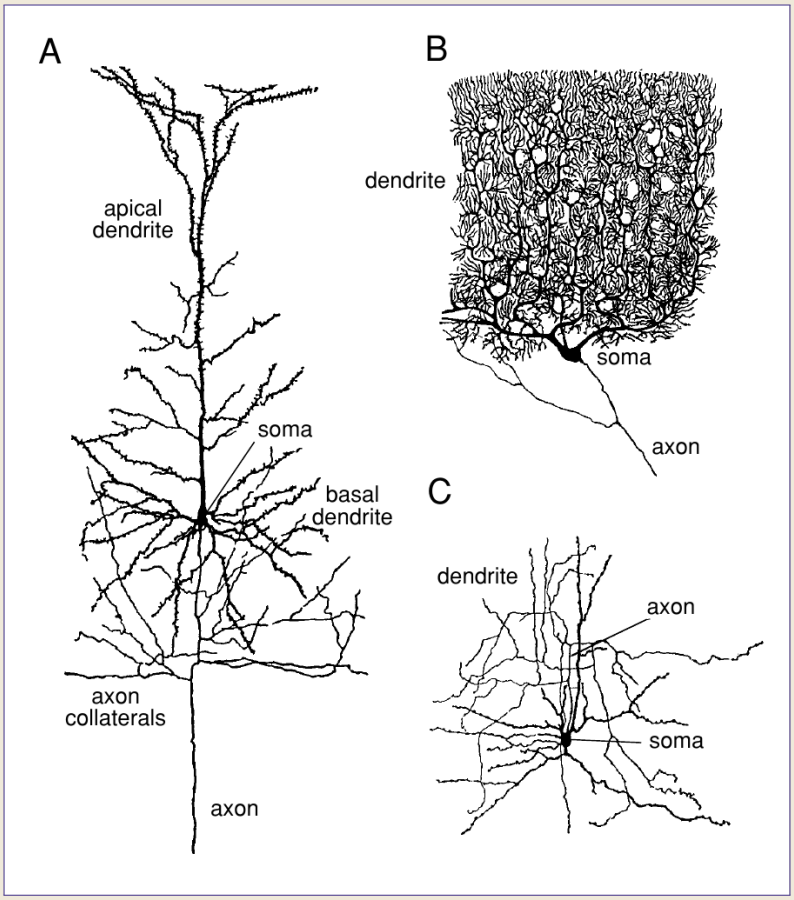
DEF · NEURON

A neuron is a nerve cell, the basic computational unit of the brain. It receives electrical inputs from other neurons, combines them, and in response produces short electrical pulses (*spikes*) that it sends on to other neurons.

Every neuron, whatever its shape, has the same three parts. The lecture gives each part a one-word job:

DEF · SOMA, AXON, DENDRITES

- Soma = the cell body. A bulb roughly 10–50 μm across that holds the nucleus.
- Axon = the output. A thin cable (0.1–10 μm thick, but up to 1 m long in humans) that carries the neuron's signal away from the soma to the axon terminals. Side branches are called axon collaterals.
- Dendrites = the input. Thin, branching extensions that collect signals from other neurons. On pyramidal cells we distinguish the apical dendrite (the long one pointing up) from the basal dendrites (near the soma).



Three neurons drawn from real cells (lecture slide 4). (A) a cortical pyramidal cell with apical and basal dendrites, soma, axon and axon collaterals; (B) a Purkinje cell of the cerebellum with its huge, dense dendritic fan; (C) a stellate cell. The shapes are very different, but each has a soma, an axon and dendrites.

WHY ARE NEURONS SHAPED LIKE TREES?

To connect, the axon of one neuron must physically touch the dendrite of another. So a neuron that listens to many others needs a big dendritic tree. The cerebellar Purkinje cell (B) receives input from *hundreds of thousands* of neurons, and its dendrites fill a huge fan. A cerebellar granule cell receives from about four neurons, and its dendritic tree is tiny [LMLF 1.1.3]. **Shape follows the number of inputs.**

> 1.2 The neuron's electricity: membrane potential, ion channels, ion pumps

The lecture says that "the dynamics of the membrane potential are determined by the flow of ions through ion channels and ion pumps". Here is what each of these words means, following [LMLF 1.1.2].

DEF · ION

An ion is an atom that carries an electric charge. Body fluids are salty water, full of ions. The two that matter most here are both positive: sodium (Na^+) and potassium (K^+).

DEF · MEMBRANE

The membrane is the thin fatty skin (a lipid bilayer) around the cell. Ions cannot pass through the fat itself, so the membrane separates two salty liquids, the *inside* and the *outside* of the neuron.

DEF · ION PUMP

An ion pump is a membrane protein that uses energy to push ions *against* their natural flow. Pumps keep K^+ concentrated *inside* the cell and Na^+ concentrated *outside*. They act like a battery charger: they build up concentration differences that other proteins can later spend.

DEF · ION CHANNEL

An ion channel is a pore-forming protein in the membrane. When it is open, it lets one specific kind of ion through (a potassium channel passes K^+, a sodium channel passes Na^+). Ion channels are tiny, about 5 nm (see the potassium channel in §5).

DEF · MEMBRANE POTENTIAL

The membrane potential V is the voltage of the inside of the cell relative to the outside, measured in millivolts (mV). At rest it is about $-70\,\text{mV}$: the inside is negative.

Why is the inside negative? Step by step:

1. The pumps make K^+ more concentrated inside than outside.
2. At rest, most of the open channels are *potassium* channels.
3. So K^+ ions leak *out* along their concentration difference.
4. Each K^+ that leaves carries a + charge out, so the inside becomes negative.
5. The result is a resting potential of about $-70\,\text{mV}$.

DEF · DEPOLARIZE / HYPERPOLARIZE

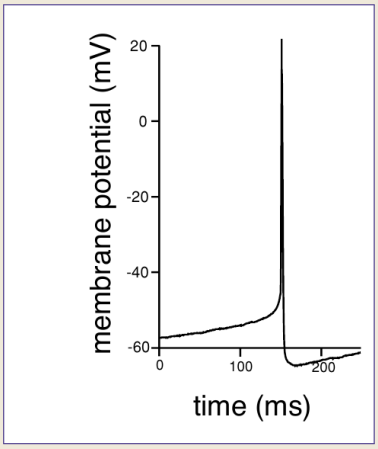
To depolarize means to make V less negative (for example $-70 \to -60\,\text{mV}$), which pushes the neuron towards firing. To hyperpolarize means to make V more negative (for example $-70 \to -80\,\text{mV}$), which pushes it away from firing. Na^+ flowing in depolarizes; K^+ flowing out hyperpolarizes.

SECTION 02

Spikes (action potentials)

DEF · SPIKE / ACTION POTENTIAL

When the membrane potential reaches a threshold, a highly nonlinear process starts and lasts about 1 ms: the voltage shoots up and comes straight back down. This brief pulse is a spike, also called an action potential. Spikes travel along the axon and provide input to the neuron's postsynaptic partners (the neurons its axon connects to).



A recorded spike (lecture slide 9). For ∼150 ms the membrane potential creeps up slowly from about −57 mV. Once it crosses threshold it jumps to about +20 mV and falls back in about 1 ms, briefly overshooting below where it started.

The mechanism, step by step [LMLF 1.1.2]:

1. Inputs depolarize the membrane a little.
2. Sodium channels are *voltage dependent*: depolarization opens some of them.
3. Na^+ rushes in, which depolarizes the membrane further and opens even more sodium channels.
4. Past the threshold (about −50 mV) this positive feedback runs away: V rises to about +40 mV in under a millisecond.
5. The sodium channels then shut (inactivate) and extra potassium channels open. K^+ flows out and V returns to rest.

EXAMPLE · SPIKE NUMBERS

Rest −70 mV, threshold ≈ −50 mV, peak ≈ +40 mV. A single spike is therefore a swing of about 110 mV in about 1 ms. The inputs that build up to it move V by only a few mV each.

STEREOTYPED, ALL-OR-NONE

Spikes have highly stereotyped temporal dynamics: every spike of a neuron looks the same. Either the threshold is crossed and a full spike happens, or it is not and nothing happens. Adrian showed in the 1920s that a stronger stimulus does *not* make bigger spikes. It makes *more* spikes per second [LMLF 1.1.2].

CENTRAL DOGMA OF NEUROSCIENCE

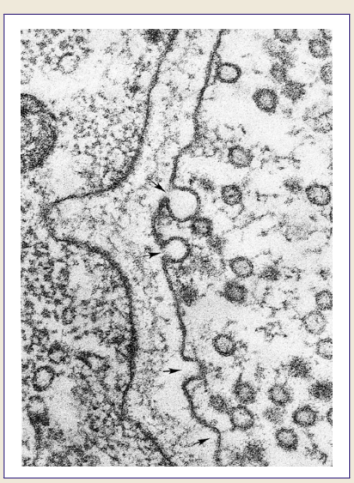
Spikes are the currency of communication in the brain. The *shape* of a spike carries no information. Only *whether* and *when* spikes occur (and therefore how many per second) carries information.

SECTION 03

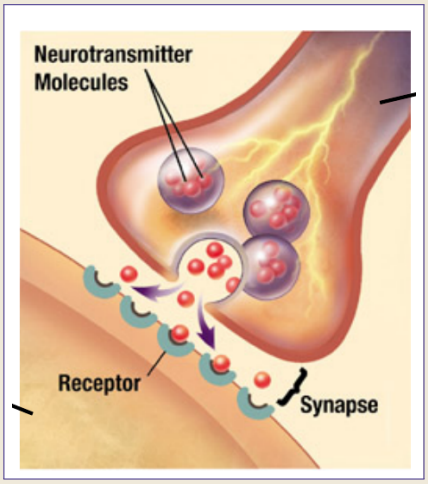
Chemical synapses

DEF · SYNAPSE, PRESYNAPTIC, POSTSYNAPTIC

A synapse is the contact point where the axon of one neuron meets (usually) a dendrite of another. The sending neuron (its axon side) is the presynaptic neuron. The receiving neuron (its dendrite side) is the postsynaptic neuron. In a chemical synapse the two are separated by a gap of about 30 nm [LMLF 1.1.3].



electron micrograph



axon (top) onto dendrite (bottom)

Chemical synapses (lecture slide 10). Left: a real synapse under the electron microscope. The small round vesicles hold neurotransmitter, and the arrows mark vesicles fusing with the membrane. Right: a cartoon of the same process.

What happens when a spike reaches a chemical synapse:

1. The spike arrives at the axon terminal of the presynaptic neuron.
2. Small bubbles (vesicles) full of neurotransmitter molecules fuse with the membrane and release their contents into the gap.
3. The neurotransmitter diffuses across the ~ 30 nm gap.
4. It binds to receptors, proteins on the postsynaptic membrane.
5. The receptors open or close ion channels. This converts the chemical signal back into an electrical signal and changes the postsynaptic membrane potential.

DEF · EXCITATORY AND INHIBITORY SYNAPSES

An excitatory synapse *depolarizes* the postsynaptic neuron (moves it towards threshold). An inhibitory synapse *hyperpolarizes* it (moves it away from threshold).

PITFALL · SLIDE WORDING

The slide says synapses "may either hyperpolarize or polarise the membrane potential". The intended words are **hyperpolarize (inhibit) or *depolarize* (excite)**. The membrane is always polarized; a synapse makes it more or less so.

DALE'S PRINCIPLE

The synaptic connections *from* a given neuron are either all excitatory or all inhibitory. A neuron releases the same neurotransmitters at all its terminals, so it has the same sign at every target. Therefore we can label whole *neurons* as excitatory or inhibitory.

PITFALL · "FROM", NOT "TO"

Dale's principle concerns the *outgoing* synapses. A neuron can *receive* both excitatory and inhibitory input; only what it *sends* has one sign. In the rate model below this means that all the weights in one *column* j of J (the synapses out of neuron j) have the same sign. The weights in a row can have mixed signs.

SECTION 04

The neuron as a computational unit: the rate model

The lecture now strips the biology down to a simple computing element.

SIMPLE VIEW OF A NEURON

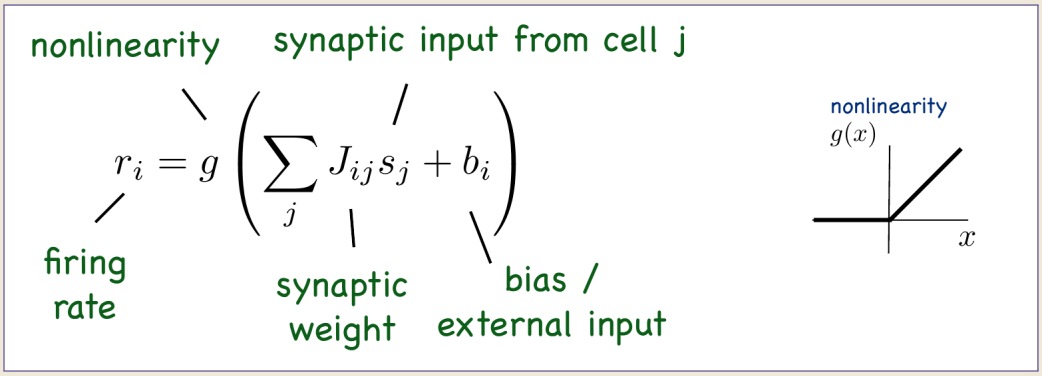
A neuron

1. receives input in the form of spikes;
2. sums these inputs linearly, with weights that reflect the strength of each synapse;
3. decides, based on this linear sum, when to produce a spike.

We usually go one step further and suppose that the neuron decides only on a firing rate, the number of spikes per second, and we ignore the exact spike times.

DEF · FIRING RATE

The firing rate r of a neuron is how many spikes it emits per second, in Hz. Example: 30 spikes in half a second $\Rightarrow r = 60\,\text{Hz}$.



The rate model as annotated in the lecture (slide 12), with the threshold-linear nonlinearity on the right.

RATE MODEL (NO DYNAMICS YET)

$$r_i = g\Big(\sum_j J_{ij}\, s_j + b_i\Big) \quad \xrightarrow{s_j = r_j} \quad r_i = g\Big(\sum_j J_{ij}\, r_j + b_i\Big)$$

r_i	firing rate of neuron i
s_j	synaptic input from cell j
J_{ij}	synaptic weight from neuron j *to* neuron i (>0 excitatory, <0 inhibitory)
b_i	bias / external input to neuron i
g	the nonlinearity (activation function)

PITFALL · INDEX ORDER

J_{ij} is the synapse **from** j **to** i: the first index is the receiver. Row i of J lists everything neuron i hears.

DEF · NONLINEARITY g

g turns the total input h into a firing rate. The lecture draws the threshold-linear function (also called ReLU):

$$g(x) = \lfloor x \rfloor = \begin{cases} x, & x > 0 \\ 0, & x \le 0. \end{cases}$$

It encodes the fact that a firing rate cannot be negative: below threshold the neuron is silent, above it the rate grows with the input. Other common choices in the book are the S-shaped *logistic* function, which also saturates, and, for small changes, the linear function $g(x) = x$ [LMLF 2.2.1].

EXAMPLE · THREE NEURONS, ONE STEP

Neuron 3 receives from neurons 1 and 2. Take $r_1 = 10\,\text{Hz}$, $r_2 = 20\,\text{Hz}$, $J_{31} = 0.5$ (excitatory), $J_{32} = -0.4$ (inhibitory), bias $b_3 = 2$, and g threshold-linear.

$$h_3 = 0.5 \cdot 10 + (-0.4) \cdot 20 + 2 = 5 - 8 + 2 = -1 \;\Rightarrow\; r_3 = g(-1) = 0\ \text{Hz}.$$

If neuron 2 slows to $r_2 = 5\,\text{Hz}$: $h_3 = 5 - 2 + 2 = 5 \Rightarrow r_3 = 5\,\text{Hz}$. Inhibition switched the neuron off, and removing most of it switched the neuron back on.

IDENTIFYING s_j WITH r_j

The input neuron i gets from cell j is assumed to be proportional to how fast j fires. Once $s_j = r_j$, the same rates appear on both sides of the equation: every neuron's output is somebody else's input. That is what makes it a *network*.

PITFALL · NO DYNAMICS YET

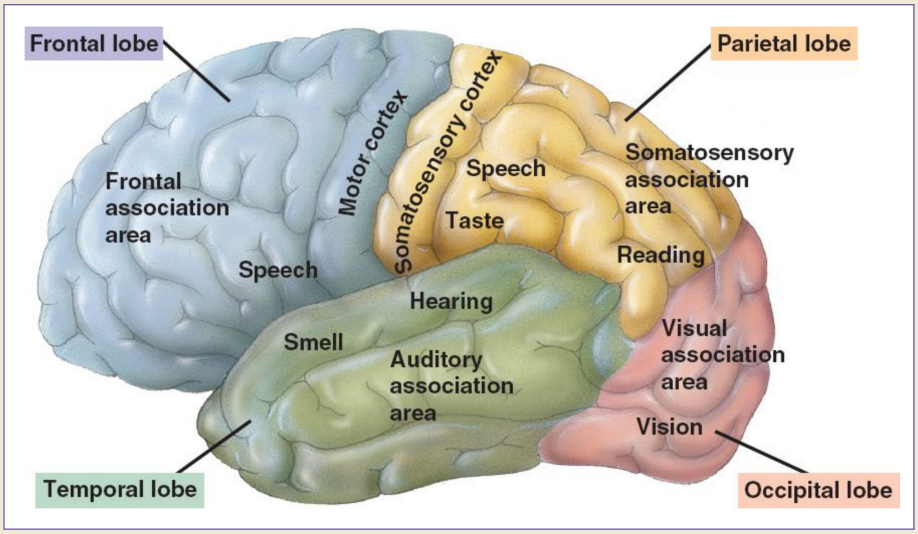
The lecture stresses: **so far there are no dynamics**. The equation says what the rate *is*, not how it *changes in time*. Time enters in the bonus hour and the TA session as $\tau \dot{r}_i = -r_i + g(\sum_j J_{ij} r_j + b_i)$ (Part B, §20).

SECTION 05

Scales in the human brain

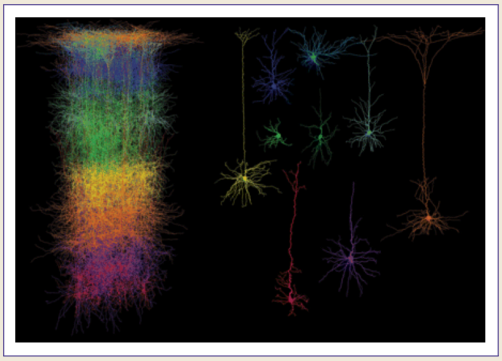
> 5.1 Length

The brain has structure on every length scale, from single molecules to the whole nervous system.

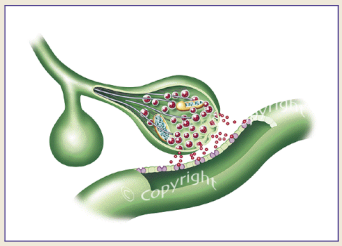


The cerebral cortex (Hebrew: *klipat ha-moach*, "the brain's peel"), the folded outer sheet of the brain (slide 14). It is divided into lobes: frontal (motor cortex, speech, association areas), parietal (somatosensory, reading), temporal (hearing, smell) and occipital (vision).

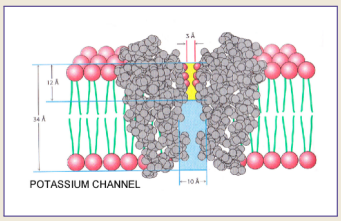
LEVEL	SIZE	WHAT IT IS
Molecules	1 nm	single proteins, e.g. an ion channel (∼5 nm)
Synapses	1 μm	one axon–dendrite contact
Neurons	10–100 μm	size of the cell body
Local networks	1 mm	a cortical column, $\sim 10^4$ neurons
Maps	1 cm	an area representing e.g. the visual field or body surface
Systems	10 cm	e.g. the visual system, the motor system
Central nervous system	1 m	brain + spinal cord



cortical column, ∼1 mm (Oberlaender . . . Sakmann 2011)



synapse, ∼1 μm



potassium channel, ∼5 nm

Three of the length scales (slide 16). In the column the colours mark neurons whose cell bodies sit in different layers.

WHERE THIS COURSE LIVES

The lecture circles two levels: local networks (1 mm) and maps (1 cm). The rate model describes networks of many neurons. It does not describe molecules or the inside of a single synapse.

> 5.2 Time

PROCESS	TIME	MEANING
Action potential	1 ms	duration of one spike
Membrane integration time	10 ms	how long the membrane "remembers" input
Synaptic integration time	1–100 ms	how long one synaptic input lasts
Short-term synaptic plasticity	0.1–1 s	synapses temporarily strengthen/weaken with use
Long-term synaptic plasticity	1–10^4 s	lasting change of synaptic strength
Long-term structural changes	10^6–10^9 s	long-term memory, learning

EXAMPLE · HOW WIDE IS THE RANGE?

From 10^{-3} s (a spike) to 10^{9} s (≈ 30 years) is a factor of 10^{12}. Lengths run from 10^{-9} m to 1 m, a factor of 10^{9}. The membrane integration time of ~ 10 ms is the τ that appears in the rate model (Part B, §20).

SECTION 06

Numbers in human and non-human brains

ANIMAL	# NEURONS	# SYNAPSES	SYNAPSES / NEURON
C. elegans (worm)	302	5000	≈ 17
Drosophila (fruit fly)	10^5	10^7	10^2
M. musculus (mouse)	10^8	10^{11}	10^3
H. sapiens (human)	10^{11}	10^{14}–10^{15}	10^3–10^4
Human cortex	2×10^{10}		
One cortical column (1 mm×1 mm)	10^4	10^7–10^8	10^3–10^4

EXAMPLE · READING THE TABLE

The last column is just synapses ÷ neurons. For a human: $10^{14}/10^{11} = 10^3$ up to $10^{15}/10^{11} = 10^4$. So a typical neuron listens to thousands of others. A single column already holds as many synapses as an entire fly brain.

SECTION 07

Why the brain is hard: challenges

- **Collective dynamics of many computational units.** Behaviour emerges from 10^{11} interacting neurons, not from any single one.
- **A vast range of temporal and spatial scales** (§5).
- **Complex structure on all scales, most of it unknown.** Which details are important? At what level of description should we model?
- **Function**: the brain computes and processes information. Understanding it means understanding *what* it computes.
- **Evolution was free to tweak properties on all scales** to achieve a function, and it often uses multiple mechanisms for the same job.

SECTION 08

Computational neuroscience

DEF · COMPUTATIONAL NEUROSCIENCE

Computational neuroscience (Hebrew: *chishuviyut atzabit*) is the study of the brain in terms of the computational and informational properties of the nervous system. The interest is often at the systems level: the collective behaviour of many neurons.

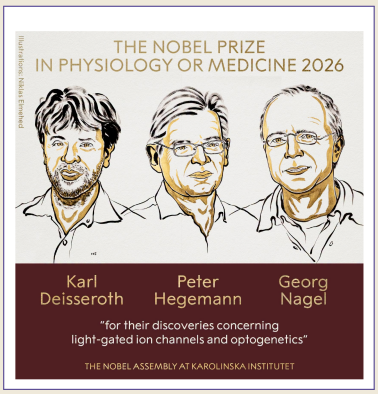
Computations the brain performs: sensory perception (discrimination, object recognition, . . .), decision making, generation of motor output, memory, learning, language and creativity.

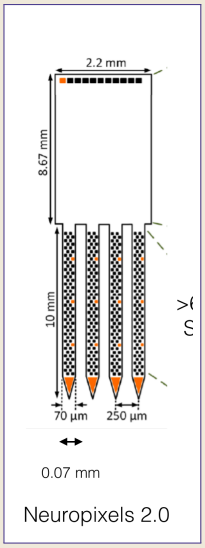
INFORMATION EVERYWHERE

Information plays a central role in all of these: it is represented (coded), extracted, discarded, relayed and stored. The "coding" half of the course title is about this.

SECTION 09

New tools in neuroscience





Left: the 2026 Nobel Prize in Physiology or Medicine (Deisseroth, Hegemann, Nagel) for light-gated ion channels and optogenetics (slide 23). Right: a Neuropixels 2.0 probe, four shanks 70 μm wide with more than 6000 recording sites (slide 24).

DEF · OPTOGENETICS

Optogenetics: neurons are genetically made to produce *light-gated ion channels*, i.e. ion channels (§1.2) that open when light shines on them. Shining light then activates or silences chosen neurons. Because a light-opened Na^+-type channel depolarizes and a Cl^-/K^+-type one hyperpolarizes, the same trick works in both directions.

DEF · CALCIUM IMAGING

Calcium imaging: neurons are made to express a molecule that glows when it binds calcium. Spikes are followed by calcium entry, so the brightness of a neuron is a proxy for its activity. This gives the firing *rates* of many neurons, but not the individual spikes [LMLF 2.2.3].

The lecture lists the technological revolutions:

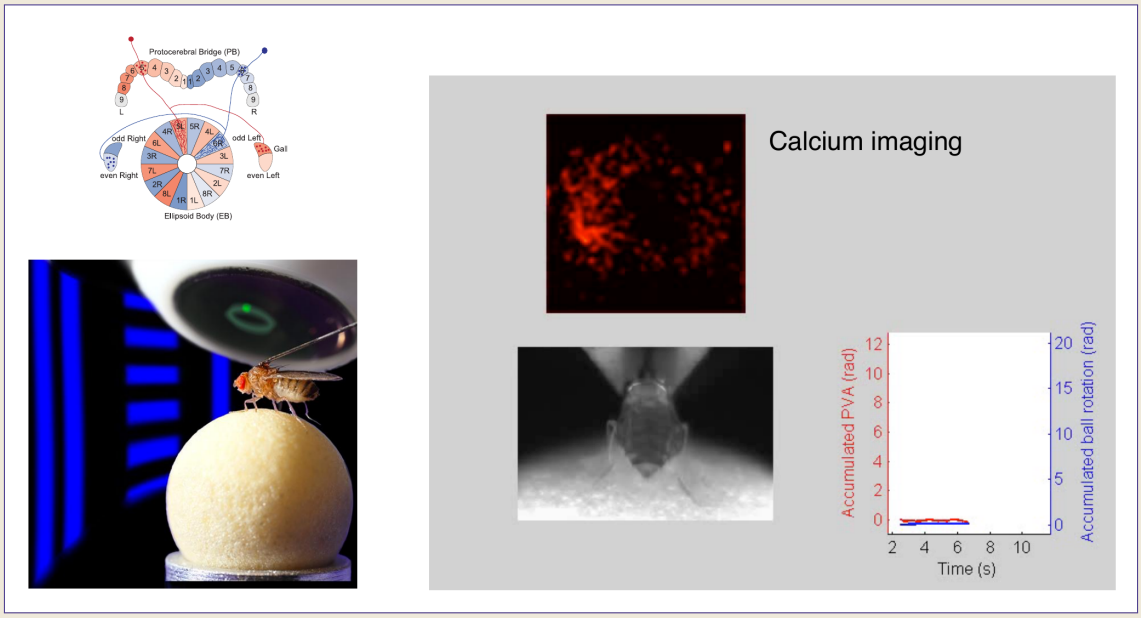
- optogenetic activation or silencing of individual neurons;
- simultaneous recordings from hundreds of neurons (silicon probes, calcium imaging);
- all of this in awake, freely behaving animals, relying heavily on genetic tools;
- anatomical tracing of connectivity (connectomics).

WHY THIS FAVOURS THE RATE MODEL

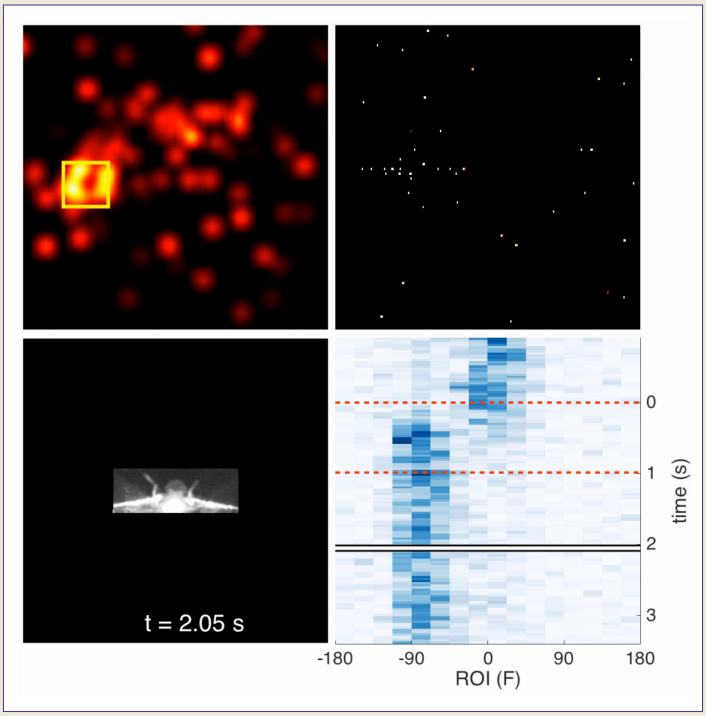
Calcium imaging reports roughly how active each neuron is, not when each spike happened. The rate model is the simplest dynamical model whose variables are exactly what these experiments measure [LMLF 2.2.3].

SECTION 10

A case study: heading in the fruit fly brain



Representation of heading in the fruit fly (slide 25; Seelig & Jayaraman, Nature 2015; Kim, Rouault, Druckmann & Jayaraman, Science 2017). Bottom left: a fly walks on an air-supported ball in front of a visual display while a microscope records from its brain. Top left: the ellipsoid body (EB), a ring of 16 wedges. Right: calcium imaging of the ring and the fly's turns on the ball.



Slide 26. Bottom right: activity of each wedge of the ring (x-axis, from $-180°$ to $180°$) over time (y-axis). At any moment only a few neighbouring wedges are active, forming a bump. The bump moves around the ring as the fly turns, and it stays in place when the fly stands still, even in the dark.

WHAT THE EXPERIMENT SHOWS

The ring of neurons works like an internal compass. The position of the activity bump on the ring *is* the fly's heading. When the fly turns, the bump moves by the same angle. The bump persists without visual input, so the network *remembers* the heading.

SECTION 11

The theoretical questions: dynamics and coding

The fly compass raises the two families of questions that give the course its name:

Dynamics

- How can a network of neurons generate activity structured as a bump?
- How can memory be sustained in the ongoing neural activity?
- How does the network respond to inputs? ("input selectivity")

Coding

- What is the best way to decode the represented heading from the spikes of these neurons?
- How precise is the neural representation?

SECTION 12

Learning

The word "learning" is used in two ways in neuroscience:

DEF · LEARNING, TWO VIEWS

- **Mechanistic**: any process that involves long-term changes in synaptic connectivity in the brain (synaptic plasticity). In the rate model, this means changes of the weights J_{ij}.
- **Functional**: shaping neural connectivity to *achieve a function*. An important distinction here is supervised learning (a teacher signal says what the right output is) versus unsupervised learning (no teacher).

Learning is covered in a separate course (*Neural Learning*, semester B), not in this one.

SECTION 13

Roadmap of the course

PART I: NEURAL NETWORK DYNAMICS (LOEWENSTEIN)	PART II: NEURAL CODING (BURAK)
Rate model (ch. 2): single neuron, network dynamics; autapse, mutual inhibition, oscillations. **Feature selectivity** (ch. 3): feedforward and recurrent networks, ring model, amplification. **Working memory** (ch. 4): line attractor, integration, ring model. **Balanced networks** (ch. 6). *Tools: linear algebra, differential equations.*	**Decoders**: maximum-likelihood and MAP decoders, discriminators, estimators (Fisher information, Cramér–Rao bound), Poisson neurons. **Efficient coding**: Shannon information, optimal nonlinearity, PCA, receptive fields in the retina. *Tools: statistics, linear algebra.*

Part B and Part C below supply exactly the two Part I tools: differential equations and linear algebra.

PART B · BONUS HOUR 1

Differential Equations: $\dot{n} = an$ and Its Relatives

SECTION 14

Why "dynamics"?

"Dynamic" means *characterized by constant change, activity or progress.* This course studies quantities that keep changing, such as firing rates, and it cares more about how they change than about the values they hold at any one moment.

ONE PATH VS. ALL PATHS

Knowing one trajectory of a system, out of infinitely many, tells you almost nothing. Understanding how *all* the trajectories behave tells you everything. **A dynamical system** is a set of differential equations used as a model of how something in the world changes.

SECTION 15

Vocabulary

DEF · DERIVATIVE, DOT NOTATION

$\frac{dn}{dt}$ is the rate of change of n at time t. We also write it $\dot{n}(t)$ or simply $\dot{n}$ (Newton's dot notation). Example: if n goes from 100 to 103 in 0.1 days, then $\dot{n} \approx 3/0.1 = 30$ per day.

DEF · DIFFERENTIAL EQUATION

A differential equation is an equation that relates an unknown *function* to its derivative, for example $\dot{n}(t) = a\,n(t)$. It is first order if only the first derivative appears, and linear if n and $\dot{n}$ appear only to the first power, without products [LMLF 2.A.1].

PITFALL · THE UNKNOWN IS A FUNCTION, NOT A NUMBER

Compare $f(x) = ax$ with $\dot{n}(t) = an(t)$. Solving for x gives *one number*. Solving the differential equation gives a *whole function* $n(t)$, i.e. a value at every time.

DEF · INITIAL CONDITION

The initial condition $n(0) = n_0$ is the value at the start. A differential equation has a whole family of solutions, and the initial condition picks out one of them.

SECTION 16

A model of an epidemic

The bonus hour builds an equation from a story. A mysterious illness spreads; you know roughly how many people get sick and how many recover each day. Let $n(t)$ be the number of sick people at time t.

EPIDEMIC MODEL

$$\tau \frac{dn}{dt} = \tau \dot{n}(t) = \beta\, n(t) - \gamma\, n(t)$$

$\beta n(t)$	new infections: more sick people ⇒ more infections (β = rate of infection)
$-\gamma n(t)$	recoveries: a fixed fraction γ of the sick recover per unit time
τ	the time constant: the time frame of these processes (e.g. days)

Simplify. Collect the terms and divide by τ:

$$\tau \dot{n} = (\beta - \gamma)\, n \iff \dot{n} = \underbrace{\frac{\beta - \gamma}{\tau}}_{a} n \iff \boxed{\dot{n}(t) = a\, n(t)}$$

The sign of a tells the story:

- $a > 0$ ($\beta > \gamma$): more people get sick than recover, so n grows.
- $a < 0$ ($\beta < \gamma$): more people recover than get sick, so n shrinks.
- $n(t_0) = 0$: nobody is sick, so nobody infects or recovers, $\dot{n} = 0$, and n stays 0 forever.

SECTION 17

Solving $\dot{n} = an$ step by step

The trick is to multiply by a cleverly chosen function so that the left-hand side becomes one single derivative.

Step 0: the product rule.

$$\frac{d}{dt}\big(f(t)\, g(t)\big) = \dot{f}(t)\, g(t) + f(t)\, \dot{g}(t).$$

Step 1: move everything to one side. $\dot{n} - an = 0$.

Step 2: multiply by e^{-at}. $\dot{n}\, e^{-at} - a\, n\, e^{-at} = 0$.

Step 3: recognise a product rule. Take $f = n$ and $g = e^{-at}$, so $\dot{g} = -ae^{-at}$. Then

$$\frac{d}{dt}(n\,e^{-at}) = \dot{n}\,e^{-at} + n\,(-ae^{-at}) = \dot{n}\,e^{-at} - a\,n\,e^{-at},$$

which is exactly the left-hand side of Step 2. So

$$\frac{d}{dt}(n(t)\,e^{-at}) = 0.$$

Step 4: integrate. A function whose derivative is 0 is a constant: $n(t)\,e^{-at} = C$.

Step 5: solve for n. $n(t) = Ce^{at}$.

Step 6: name the constant. At $t = 0$: $n(0) = Ce^0 = C$. So $C = n_0$:

SOLUTION

$$\dot{n} = an, \quad n(0) = n_0 \qquad \Longrightarrow \qquad \boxed{n(t) = n_0\,e^{at}}$$

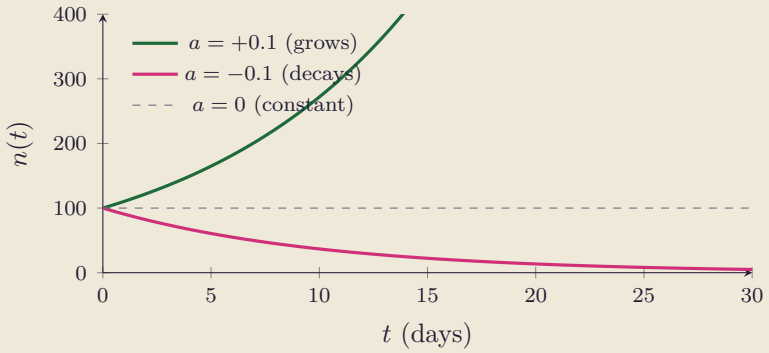
Check by plugging in. $\frac{d}{dt}(n_0 e^{at}) = a\,n_0 e^{at} = a\,n(t)$. ✓

EXAMPLE · NUMBERS FOR THE EPIDEMIC

Start with $n_0 = 100$ sick people, time in days.

DAY t	$a = +0.1$: $100e^{0.1t}$	$a = -0.1$: $100e^{-0.1t}$
0	100	100
7	201	49.7
14	406	24.7
21	817	12.2
30	2009	5.0

With $a = 0.1$ the number doubles every $\ln 2/a \approx 6.9$ days. With $a = -0.1$ it halves every 6.9 days. Same speed, opposite directions.



$n(t) = n_0 e^{at}$ for three values of a. The sign of a alone decides between growth and decay.

PITFALL · LIMITS OF THE MODEL

Every model is imperfect. This one:

1. assumes an **infinite population**: with $a > 0$, n grows forever (on day 100 we would have $100e^{10} \approx 2.2$ million sick);
2. is **continuous**: n need not be an integer (3.1 sick people), and with $a < 0$ it never reaches exactly 0, it only approaches it.

SECTION 18

When we cannot solve: three other ways to analyse

Most equations in the course cannot be solved by a formula. The bonus hour names three alternatives:

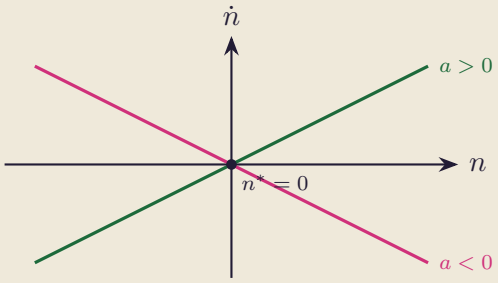
1. **Numerical methods.** Use the derivative to step forward in time on a computer. The simplest is the Euler method [LMLF 2.A.1]: $n(t + dt) \approx n(t) + \dot{n}(t)\, dt$.
2. **Fixed points.** Ask questions about the derivative instead of the function.
3. **Geometry.** Use the sign of the derivative: $\dot{n} > 0$ means n is increasing, $\dot{n} < 0$ means it is decreasing.

EXAMPLE · EULER BY HAND

$\dot{n} = 0.1n$, $n(0) = 100$, step $dt = 1$ day: $n(1) \approx 100 + 0.1 \cdot 100 \cdot 1 = 110$, $n(2) \approx 110 + 11 = 121$, $n(3) \approx 133.1$. The exact values are 110.5, 122.1, 135.0. A smaller dt gives a closer match.

DEF · FIXED POINT

A fixed point is a state where the dynamics stand still: $\dot{n} = 0$. If the system starts there, it stays there. For $\dot{n} = an$ with $a \neq 0$ the only fixed point is $n^* = 0$ (no sick people ⇒ no change).



The geometric view: plot $\dot{n}$ against n. Where the line is above the axis n grows, and where it is below n shrinks. For $a < 0$ every state flows towards 0; for $a > 0$ every state flows away from 0.

The bonus hour asks how the system *behaves around* a fixed point. That question has a name.

DEF · CONVERGE, DIVERGE

A solution converges to n^* if $n(t) \to n^*$ as $t \to \infty$: it gets as close as you like and stays there. It diverges if $|n(t)|$ grows without bound. Example: $n(t) = 100e^{-0.1t}$ converges to 0 ($100 \to 36.8 \to 13.5 \to \dots$ at $t = 0, 10, 20$); $n(t) = 100e^{0.1t}$ diverges ($100 \to 272 \to 739 \to \dots$).

DEF · STABLE AND UNSTABLE FIXED POINT

Start the system *near* the fixed point, not exactly on it.

- The fixed point is stable if every such start converges back to it.
- It is unstable if some starts arbitrarily close to it move away from it.

For $\dot{n} = an$: if $a < 0$, $n(t) = n_0 e^{at} \to 0$ from any n_0, so $n^* = 0$ is stable. If $a > 0$, even $n_0 = 0.001$ grows like $0.001e^{at}$, so $n^* = 0$ is unstable.

A BALL ON A LANDSCAPE

A stable fixed point is a ball resting at the bottom of a valley: nudge it and it rolls back. An unstable one is a ball balanced on a hilltop: it is at rest too, but the smallest nudge sends it away. In the plot above, the line crossing zero going down ($a < 0$) means stable; going up ($a > 0$) means unstable.

SECTION 19

Adding outside input: $\dot{n} = an + I(t)$

Now let infected travellers enter the country at a rate $I(t)$ (people per unit time), and take $a < 0$. The equation becomes

$$\dot{n}(t) = a\, n(t) + I(t).$$

SOLUTION WITH INPUT

$$\boxed{n(t) = n_0\, e^{at} + \int_0^t I(\tilde{t})\, e^{a(t-\tilde{t})}\, d\tilde{t}}$$

Where it comes from (same trick as before) [LMLF 2.A.1]:

1. Rewrite: $\dot{n} - an = I$. Multiply by e^{-at}: $\frac{d}{dt}(n\, e^{-at}) = I(t)\, e^{-at}$.
2. Integrate from 0 to t: $n(t)e^{-at} - n_0 = \int_0^t I(\tilde{t})\, e^{-a\tilde{t}}\, d\tilde{t}$.
3. Multiply by e^{at}: $n(t) = n_0 e^{at} + \int_0^t I(\tilde{t})\, e^{a(t-\tilde{t})}\, d\tilde{t}$.

READING THE FORMULA

- $n_0 e^{at}$: the people who were sick at the start, whose number decays (for $a < 0$) exactly as before.
- $\int_0^t I(\tilde{t}) e^{a(t-\tilde{t})} d\tilde{t}$: every earlier arrival, weighted by how long ago it happened. People who

arrived at time $\tilde{t}$ have had $t-\tilde{t}$ time to recover (or to spread the illness), so their contribution today is $I(\tilde{t})$ multiplied by $e^{a(t-\tilde{t})}$.

If a big group arrives at some time t_1, it shows up as a jump at t_1 that then fades with the factor $e^{a(t-t_1)}$.

EXAMPLE · CONSTANT INPUT: A NEW BALANCE POINT

Take $a=-0.1$/day, a constant $I=5$ people/day, and $n_0=100$. The integral is $\int_0^t 5\,e^{-0.1(t-\tilde{t})}d\tilde{t} = 50\,(1-e^{-0.1t})$, so

$$n(t) = 100e^{-0.1t} + 50(1-e^{-0.1t}) = 50 + 50\,e^{-0.1t}.$$

t (days)	0	7	14	30	60
$n(t)$	100	74.8	62.3	52.5	50.1

The number settles at $n^*=50$, which is the fixed point: $\dot{n} = -0.1\cdot 50 + 5 = 0$. In general a constant input gives $n^* = -I/a$ [LMLF eq. 2.64]. Arrivals (5/day) exactly balance recoveries (0.1×50/day).

DEF · CONVOLUTION

An integral of this form is a convolution:

$$(f * g)(t) = \int_0^t f(\tilde{t})\,g(t-\tilde{t})\,d\tilde{t}.$$

Our integral is $I * e^{at}$. It sums up the contributions of the input at all past times, each one weighted ("modulated") by e^{at} evaluated at how long ago it happened. The limits $[0,t]$ sometimes change (e.g. $(-\infty,t]$).

SECTION 20

The link to the rate model

The rate model of the course has exactly this shape:

RATE MODEL WITH DYNAMICS

$$\tau\,\dot{r} = -r + g(h), \qquad h = Jr + h^{\text{ext}}$$

For a network of N neurons: $\tau\dot{r}_i = -r_i + g\Big(\sum_{j=1}^{N} J_{ij}r_j + h_i^{\text{ext}}\Big)$.

Compare it with $\dot{n} = an + I$ term by term:

EPIDEMIC	RATE MODEL
n = number sick	r = firing rate
$a\,n$ with $a<0$	$-r/\tau$: decay with $a=-1/\tau$
input $I(t)$	$g(h(t))/\tau$

DECAY TOWARDS SILENCE

The $-r$ term is a decaying exponential. Without stimulation, a neuron's rate returns to zero on the time scale τ. Input through $g(h)$ pushes it up again. The difference from the epidemic is that h depends on r itself (through J), so the "input" is fed back by the network. This is what makes network dynamics interesting.

EXAMPLE · WHAT τ MEANS, WITH NUMBERS

A single neuron with a constant input h switched on at $t = 0$, starting from $r(0) = 0$. Plugging $a = -1/\tau$ and the constant input $g(h)/\tau$ into the solution with input gives [LMLF eq. 2.7]

$$r(t) = g(h)(1 - e^{-t/\tau}).$$

With $\tau = 10\,\text{ms}$ (the membrane integration time of §5):

t (ms)	0	5	10	20	30	50
$r/g(h)$	0	0.39	0.63	0.86	0.95	0.99

τ is the time to reach $1 - 1/e \approx 63\%$ of the final rate. The neuron does not respond instantly; it follows its input with a lag of about τ.

PITFALL · SHORTHAND

We usually write $r, \dot{r}$ for $r(t), \dot{r}(t)$. Also, $g(h(r(t)))$ looks scary, but in practice the terms are written out explicitly.

PART C · TA SESSION 1

Linear Systems: Solving $\dot{\vec{x}} = M\vec{x}$

SECTION 21

Why linear systems?

The TA session solves the rate model for N neurons in the simplest case, a linear activation function $g(h) = h$. Then (with $\tau = 1$ and no external input) the rate equation becomes

$$\dot{r}_i = -r_i + \sum_j J_{ij} r_j \qquad \Longleftrightarrow \qquad \dot{\vec{r}} = (J - I)\,\vec{r},$$

a system of the form $\dot{\vec{x}} = M\vec{x}$ with the matrix $M = J - I$. So we need to know how to solve $\dot{\vec{x}} = M\vec{x}$.

SECTION 22

From one equation to many

One equation. $\dot{x} = ax \Rightarrow x(t) = Ce^{at}$ (Part B).

Many separate equations. $\dot{x}_i = a_i x_i$ for $i = 1, \ldots, N$. Each equation contains only its own variable, so we solve each one separately: $x_i(t) = C_i e^{a_i t}$. Easy.

Coupled equations. Now mix the variables:

$$\dot{x}_1 = 3x_1 + 2x_2, \qquad \dot{x}_2 = x_1 - x_2.$$

DEF · COUPLED

Equations are coupled when the rate of change of one variable depends on the others. Here $\dot{x}_1$ depends on x_2 and $\dot{x}_2$ depends on x_1, so neither can be solved alone.

THE PLAN

As with high-school simultaneous equations, we want to rearrange the equations so that each one involves only itself. Linear algebra does this: in the basis of the eigenvectors the equations decouple.

Matrix form.

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 3 & 2 \\ 1 & -1 \end{pmatrix}}_{M} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \qquad \Longleftrightarrow \qquad \dot{\vec{x}} = M\vec{x}.$$

SECTION 23

Worked example: $M = \left(\begin{smallmatrix} 3 & 2 \\ 1 & -1 \end{smallmatrix}\right)$

> 23.1 Step 1: eigenvalues

DEF · EIGENVALUE, RIGHT EIGENVECTOR

λ is an eigenvalue of M with (right) eigenvector $\vec{u} \neq 0$ if $M\vec{u} = \lambda\vec{u}$: M only stretches $\vec{u}$, it does not turn it.

Solve $\det(M - \lambda I) = 0$:

$$\det \begin{pmatrix} 3-\lambda & 2 \\ 1 & -1-\lambda \end{pmatrix} = (3-\lambda)(-1-\lambda) - 2 = \lambda^2 - 2\lambda - 5 = 0 \;\Rightarrow\; \lambda = \frac{2 \pm \sqrt{4+20}}{2} = 1 \pm \sqrt{6}.$$

$$\lambda_1 = 1 - \sqrt{6} \approx -1.449, \qquad \lambda_2 = 1 + \sqrt{6} \approx +3.449.$$

> 23.2 Step 2: right eigenvectors

Write $\vec{u} = (x, 1)^\top$. The first row of $M\vec{u} = \lambda\vec{u}$ reads $3x + 2 = \lambda x$, so $x = \dfrac{2}{\lambda - 3}$.

- $\lambda_1 = 1-\sqrt{6}$: $x = \dfrac{2}{-2-\sqrt{6}} = \dfrac{2(\sqrt{6}-2)}{(-2-\sqrt{6})(\sqrt{6}-2)} = \dfrac{2(\sqrt{6}-2)}{-2} = 2-\sqrt{6}$. So $\vec{u}_1 = \begin{pmatrix} 2-\sqrt{6} \\ 1 \end{pmatrix} \approx \begin{pmatrix} -0.449 \\ 1 \end{pmatrix}$.
- $\lambda_2 = 1+\sqrt{6}$: in the same way, $\vec{u}_2 = \begin{pmatrix} 2+\sqrt{6} \\ 1 \end{pmatrix} \approx \begin{pmatrix} 4.449 \\ 1 \end{pmatrix}$.

Check $\vec{u}_1$:

$$\begin{pmatrix} 3 & 2 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 2-\sqrt{6} \\ 1 \end{pmatrix} = \begin{pmatrix} 6 - 3\sqrt{6} + 2 \\ 2 - \sqrt{6} - 1 \end{pmatrix} = \begin{pmatrix} 8 - 3\sqrt{6} \\ 1 - \sqrt{6} \end{pmatrix} = (1-\sqrt{6}) \begin{pmatrix} 2-\sqrt{6} \\ 1 \end{pmatrix}. \checkmark$$

(Indeed $(1-\sqrt{6})(2-\sqrt{6}) = 2 - \sqrt{6} - 2\sqrt{6} + 6 = 8 - 3\sqrt{6}$.)

> 23.3 Step 3: left eigenvectors

What if we multiply from the *other* side, using the same vector as a row?

$$(2-\sqrt{6},\ 1) \begin{pmatrix} 3 & 2 \\ 1 & -1 \end{pmatrix} = (7 - 3\sqrt{6},\ 3 - 2\sqrt{6}).$$

Is this a multiple of $(2-\sqrt{6},\ 1)$? It would have to be $(3-2\sqrt{6}) \cdot (2-\sqrt{6},\ 1) = (18 - 7\sqrt{6},\ 3 - 2\sqrt{6})$, and the first entries do not match. **Not an eigenvector from the left.**

DEF · LEFT EIGENVECTOR

A row vector $\vec{w}^\top$ is a left eigenvector of M if $\vec{w}^\top M = \lambda \vec{w}^\top$ (equivalently, $\vec{w}$ is a right eigenvector of $M^\top$). Left and right eigenvectors have the *same* eigenvalues, but for a non-symmetric matrix they are different vectors.

Write $\vec{w}^\top = (y,\, 1)$. The first column of $\vec{w}^\top M = \lambda \vec{w}^\top$ reads $3y + 1 = \lambda y$, so $y = \dfrac{1}{\lambda - 3}$:

$$\lambda_1 :\ \vec{w}_1^\top = \left(1 - \tfrac{\sqrt{6}}{2},\ 1\right) \approx (-0.225,\, 1), \qquad \lambda_2 :\ \vec{w}_2^\top = \left(1 + \tfrac{\sqrt{6}}{2},\ 1\right) \approx (2.225,\, 1).$$

PITFALL · WHEN DO LEFT AND RIGHT EIGENVECTORS COINCIDE?

For a **symmetric** matrix ($M^\top = M$), left and right eigenvectors are the same, and the eigenvectors can be chosen orthogonal ("orthogonally diagonalizable"). The TA notes' footnote gives the more general condition $M^\top M = M M^\top$ (a *normal* matrix). Our M is not symmetric ($M_{12} = 2 \neq 1 = M_{21}$), so its left and right eigenvectors differ. The sentence in the TA notes "…and the matrix is orthogonal diagonalizable. In this case it isn't" means: *a symmetric matrix would be orthogonally diagonalizable; this one is not.*

> 23.4 Step 4: diagonalize

Put the right eigenvectors into the columns of U and the eigenvalues on the diagonal of D:

$$U = \begin{pmatrix} 2 - \sqrt{6} & 2 + \sqrt{6} \\ 1 & 1 \end{pmatrix}, \qquad D = \begin{pmatrix} 1 - \sqrt{6} & 0 \\ 0 & 1 + \sqrt{6} \end{pmatrix}, \qquad \boxed{M = U D U^{-1}}.$$

The rows of U^{-1} are the left eigenvectors, each scaled so that $\vec{w}_k^\top \vec{u}_k = 1$ [LMLF App. 2.A.2]. Check:

$$\vec{w}_1^\top \vec{u}_1 = (1 - \tfrac{\sqrt{6}}{2})(2 - \sqrt{6}) + 1 = 6 - 2\sqrt{6}, \qquad \vec{w}_2^\top \vec{u}_2 = 6 + 2\sqrt{6}, \qquad \vec{w}_1^\top \vec{u}_2 = (1 - \tfrac{\sqrt{6}}{2})(2 + \sqrt{6}) + 1 = 0.$$

So

$$U^{-1} = \begin{pmatrix} (1 - \frac{\sqrt{6}}{2},\ 1)/(6 - 2\sqrt{6}) \\ (1 + \frac{\sqrt{6}}{2},\ 1)/(6 + 2\sqrt{6}) \end{pmatrix}.$$

WHY LEFT EIGENVECTORS ARE USEFUL

A left eigenvector of one eigenvalue is perpendicular to the right eigenvectors of all the other eigenvalues ($\vec{w}_1^\top \vec{u}_2 = 0$ above). So taking the dot product with $\vec{w}_k$ "filters out" all components except the k-th one. That is how we read off the coordinates of $\vec{x}$ in the eigenbasis.

PITFALL · THE TA NOTES' $U^{-1}DU$

The TA notes write $U^{-1} D U = M$ with the *left*-eigenvector matrix called U and the right-eigenvector matrix called U^{-1}. The order (right eigenvectors, then D, then left eigenvectors) is correct, but **the left eigenvectors must be divided by** $6 \mp 2\sqrt{6}$. Without this scaling the two matrices are not inverses of each other, and the product is not M. In these notes, U always means the matrix of right eigenvectors, as in the book: $M = U D U^{-1}$.

> 23.5 Step 5: decouple and solve

Write $\vec{x}$ in the eigenbasis: $\vec{x}(t) = \tilde{x}_1(t)\,\vec{u}_1 + \tilde{x}_2(t)\,\vec{u}_2$, i.e. $\tilde{\vec{x}} = U^{-1}\vec{x}$. Then

$$\dot{\tilde{\vec{x}}} = U^{-1}\dot{\vec{x}} = U^{-1}M\vec{x} = U^{-1}(UDU^{-1})\vec{x} = D\,\tilde{\vec{x}}.$$

D is diagonal, so the equations are decoupled:

DECOUPLED EQUATIONS

$$\dot{\tilde{x}}_1 = (1-\sqrt{6})\,\tilde{x}_1 \Rightarrow \tilde{x}_1(t) = C_1 e^{(1-\sqrt{6})t}, \qquad \dot{\tilde{x}}_2 = (1+\sqrt{6})\,\tilde{x}_2 \Rightarrow \tilde{x}_2(t) = C_2 e^{(1+\sqrt{6})t}.$$

$$\vec{x}(t) = C_1 e^{\lambda_1 t}\,\vec{u}_1 + C_2 e^{\lambda_2 t}\,\vec{u}_2, \qquad C_k = \tilde{x}_k(0) = \frac{\vec{w}_k^{\top}\vec{x}(0)}{\vec{w}_k^{\top}\vec{u}_k}.$$

EXAMPLE · START AT $\vec{x}(0) = (1, 0)$

$$C_1 = \frac{1 - \sqrt{6}/2}{6 - 2\sqrt{6}} = -\frac{1}{2\sqrt{6}} \approx -0.204, \qquad C_2 = \frac{1 + \sqrt{6}/2}{6 + 2\sqrt{6}} = \frac{1}{2\sqrt{6}} \approx 0.204.$$

Check at $t = 0$: $\frac{1}{2\sqrt{6}}\big[-(2-\sqrt{6}, 1) + (2+\sqrt{6}, 1)\big] = \frac{1}{2\sqrt{6}}(2\sqrt{6},\, 0) = (1, 0)$. ✓

$$\vec{x}(t) = \frac{1}{2\sqrt{6}}\Big[\, e^{(1+\sqrt{6})t}\begin{pmatrix} 2+\sqrt{6} \\ 1 \end{pmatrix} - e^{(1-\sqrt{6})t}\begin{pmatrix} 2-\sqrt{6} \\ 1 \end{pmatrix}\Big].$$

t	$\lvert\tilde{x}_1\rvert = 0.204\,e^{-1.449t}$	$\tilde{x}_2 = 0.204\,e^{3.449t}$	$x_1(t)$	$x_2(t)$
0	0.204	0.204	1.00	0.00
0.25	0.142	0.484	2.22	0.34
0.5	0.099	1.145	5.14	1.05
1	0.048	6.43	28.6	6.38
2	0.011	202	900	202

The $\vec{u}_1$ component *dies out*, while the $\vec{u}_2$ component *explodes*. After a short while $\vec{x}(t)$ points almost exactly along $\vec{u}_2 = (4.449, 1)$: at $t = 2$, $x_1/x_2 = 900/202 \approx 4.45$.

SECTION 24

Vocabulary: fixed points and stability in two dimensions

The words from the bonus hour (Part B: fixed point, converge, diverge, stable, unstable) carry over to vectors. Here they are in two dimensions.

DEF · STATE, TRAJECTORY, PHASE PORTRAIT

The state is the vector $\vec{x} = (x_1, x_2)$, a point in the plane. As time runs, the state moves and traces a curve, the trajectory. A phase portrait is a drawing of many trajectories at once in

the (x_1, x_2) plane, with arrows for the direction of motion. It shows how the system behaves from *every* starting point.

DEF · FIXED POINT OF $\dot{\vec{x}} = M\vec{x}$

A fixed point is a state where $\dot{\vec{x}} = \vec{0}$, i.e. $M\vec{x}^* = \vec{0}$. The origin $\vec{x}^* = \vec{0}$ is always one. If M is invertible (no eigenvalue equal to 0) it is the *only* one. Example: for $M = \begin{pmatrix} 3 & 2 \\ 1 & -1 \end{pmatrix}$, $\det M = -5 \neq 0$, so $\vec{0}$ is the only fixed point.

DEF · CONVERGE, DIVERGE, STABLE, UNSTABLE (VECTORS)

- The system converges to $\vec{0}$ if $\vec{x}(t) \to \vec{0}$, i.e. every component goes to 0. It diverges if the length $\|\vec{x}(t)\|$ grows without bound.
- The fixed point $\vec{0}$ is stable if trajectories that start near it converge to it, and unstable if some start arbitrarily close to it and move away.

FOR LINEAR SYSTEMS, "NEAR" DOES NOT MATTER

If $\vec{x}(t)$ is a solution of $\dot{\vec{x}} = M\vec{x}$, so is $c\,\vec{x}(t)$ for any number c (multiply both sides by c). So a start at $(0.001, 0)$ behaves exactly like a start at $(1, 0)$, only scaled down by 1000. Hence, for linear systems, stable $\iff$ converges from every start, and unstable $\iff$ diverges from some start.

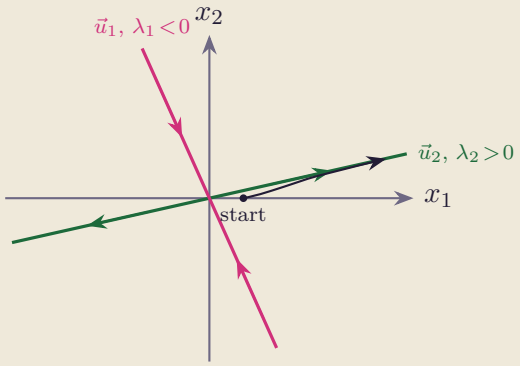
EXAMPLE · TWO DIAGONAL SYSTEMS

$M = \mathrm{Diag}(-1, -2)$, start $(1, 1)$: the equations are already decoupled, $\vec{x}(t) = (e^{-t}, e^{-2t})$. At $t = 0, 1, 2$: $(1, 1)$, $(0.37, 0.14)$, $(0.14, 0.02)$. It converges: $\vec{0}$ is **stable**.

$M = \mathrm{Diag}(1, -1)$, start $(1, 1)$: $\vec{x}(t) = (e^{t}, e^{-t})$: $(1, 1)$, $(2.7, 0.37)$, $(7.4, 0.14)$. The second component dies but the first explodes: it diverges, so $\vec{0}$ is **unstable**.

DEF · SADDLE

A saddle is a fixed point that attracts along some directions and repels along others: one eigenvalue negative, one positive. The name comes from a horse's saddle: from front to back it curves up, from side to side it curves down. A saddle is **unstable**, since almost every start eventually leaves along the repelling direction. $\mathrm{Diag}(1, -1)$ above is a saddle, and so is our worked example.



Phase portrait of the example (a *saddle*). Along $\vec{u}_1$ the state shrinks towards 0; along $\vec{u}_2$ it grows. A typical starting point (black) is first pulled in along $\vec{u}_1$ and then pushed out along $\vec{u}_2$.

SECTION 25

What the eigenvalues tell us: stability

THE WHOLE STORY IS IN THE EIGENVALUES

In the standard basis, x_1 and x_2 are linear combinations of $\tilde{x}_1$ and $\tilde{x}_2$, which we already know. So to know whether the system *converges* or *diverges*, it is enough to look at the signs of the eigenvalues; we do not need the eigenvectors.

STABILITY OF $\dot{\vec{x}} = M\vec{x}$

- **All eigenvalues negative** ⇒ every component $e^{\lambda_k t} \to 0$: from *any* start, $\vec{x}(t) \to \vec{0}$. The fixed point $\vec{0}$ is stable.
- **At least one eigenvalue positive** ⇒ that component grows like $e^{\lambda t}$: the system diverges from almost every start.

$$\boxed{\text{converges to } \vec{0} \text{ from every start} \iff \text{all eigenvalues are negative}}$$

(If the eigenvalues are complex numbers, read "negative" as "negative real part" [LMLF App. 2.A.2].)

PITFALL · CORRECTIONS TO THE TA NOTES

- The notes say "*both of them just increase over time*". That is not true for this M: $\lambda_1 = 1 - \sqrt{6} \approx -1.45$ is **negative**, so $\tilde{x}_1$ *decays*. The system still diverges, but only because $\lambda_2 > 0$.
- The notes say "*the system diverges if and only if all of the eigenvalues are negative*". This is a typo. The correct statement is **converges if and only if all eigenvalues are negative**, or equivalently **diverges if and only if at least one eigenvalue is positive**.
- "Diverges from almost every start": if you start *exactly* on the line of $\vec{u}_1$, then $C_2 = 0$ and you slide into $\vec{0}$. Any tiny deviation off that line gets amplified.

SECTION 26

The 2×2 shortcut: trace and determinant

For two neurons we do not even need the eigenvalues themselves. Two facts from linear algebra:

$$\mathrm{Tr}(M) = \sum_i \lambda_i = \lambda_1 + \lambda_2, \qquad \mathrm{Det}(M) = \prod_i \lambda_i = \lambda_1 \lambda_2.$$

The argument, step by step.

1. If both eigenvalues are negative, then their sum is negative: $\mathrm{Tr} < 0$.
2. If both eigenvalues are negative, then their product is positive (negative $\times$ negative): $\mathrm{Det} > 0$.
3. So if $\mathrm{Tr} \geq 0$ or $\mathrm{Det} \leq 0$, the eigenvalues are *not* both negative, and the system does not converge.
4. Conversely, $\mathrm{Det} > 0$ means the two eigenvalues have the same sign (or are complex conjugates with the same real part $\mathrm{Tr}/2$), and then $\mathrm{Tr} < 0$ makes that sign negative [LMLF App. 2.A.2].

2×2 STABILITY TEST

$$\dot{\vec{x}} = M\vec{x} \text{ converges to } \vec{0} \iff \boxed{\mathrm{Tr}(M) < 0 \quad \text{and} \quad \mathrm{Det}(M) > 0}$$

EXAMPLE · THREE MATRICES, NO EIGENVALUES NEEDED

M	TR	DET	VERDICT
$\begin{pmatrix} 3 & 1 \\ -4 & -1 \end{pmatrix}$	2	$-3 - 2 = -5$	Det < 0: one positive, one negative eigenvalue $\Rightarrow$ diverges
$\begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix}$	-3	2	Tr < 0, Det $> 0 \Rightarrow$ converges
$\begin{pmatrix} -2 & 1 \\ -4 & -1 \end{pmatrix}$	-3	$2 - 3 = -1$	Det $< 0 \Rightarrow$ diverges, *even though* Tr < 0

For the first one: $\lambda_1 + \lambda_2 = (1 - \sqrt{6}) + (1 + \sqrt{6}) = 2$ and $\lambda_1 \lambda_2 = 1 - 6 = -5$, matching Tr and Det. ✓

PITFALL · A NEGATIVE TRACE IS NOT ENOUGH

The third matrix shows why we need *both* conditions: its eigenvalues are $\frac{-3 \pm \sqrt{13}}{2} \approx 0.30$ and -3.30. They sum to a negative number, but one of them is positive.

Further reading from the TA notes: Appendix 2.A.2 of the book [LMLF] gives the same derivation for N dimensions: diagonalize $M = UDU^{-1}$, decouple into $\dot{\tilde{x}}_k = \mu_k \tilde{x}_k$, and the dynamics converge to 0 if all the μ_k are negative (real part).

WEEK 01

Practice Problems and Summary

SECTION 27

Practice problems with full solutions

EXAMPLE · P1: RATE MODEL

Two neurons with threshold-linear g: $J = \begin{pmatrix} 0 & -1 \\ 2 & 0 \end{pmatrix}$, $b = (3, 1)$, current rates $r = (2, 4)$. Compute $g(\sum_j J_{ij} r_j + b_i)$ for both neurons. Which neuron is inhibitory, by Dale's principle?

Solution. $h_1 = 0 \cdot 2 + (-1) \cdot 4 + 3 = -1 \Rightarrow g = 0$. $h_2 = 2 \cdot 2 + 0 \cdot 4 + 1 = 5 \Rightarrow g = 5$. The synapses *from* neuron 2 are column 2 of J: $J_{12} = -1 < 0$, so neuron 2 is inhibitory. Column 1 has $J_{21} = 2 > 0$, so neuron 1 is excitatory.

EXAMPLE · P2: SOLVING $\dot{n} = an$

A population of bacteria obeys $\dot{n} = 0.3\, n$ (per hour) with $n(0) = 1000$. When does it reach 8000?

Solution. $n(t) = 1000 e^{0.3t} = 8000 \Rightarrow e^{0.3t} = 8 \Rightarrow t = \ln 8 / 0.3 = 3 \ln 2 / 0.3 \approx 6.93\,\text{h}$. Equivalently, it doubles every $\ln 2 / 0.3 \approx 2.31\,\text{h}$, and $8 = 2^3$ means three doublings.

EXAMPLE · P3: FIXED POINT WITH INPUT

$\dot{n} = -0.5\, n + 4$, $n(0) = 0$. Find the fixed point and $n(t)$.

Solution. Fixed point: $0 = -0.5 n^* + 4 \Rightarrow n^* = 8$. Using the solution with input (with $a = -0.5$, $I = 4$, $n_0 = 0$): $n(t) = \int_0^t 4 e^{-0.5(t-\tilde{t})} d\tilde{t} = 8(1 - e^{-0.5t})$. Check: $n(0) = 0$, $n(\infty) = 8$, and $\dot{n} = 4 e^{-0.5t} = -0.5 \cdot 8(1 - e^{-0.5t}) + 4$. ✓

EXAMPLE · P4: TIME CONSTANT

A neuron with $\tau = 20\,\text{ms}$ receives a constant input from $t = 0$, with $g(h) = 50\,\text{Hz}$ and $r(0) = 0$. What is r at $t = 20\,\text{ms}$ and $t = 60\,\text{ms}$?

Solution. $r(t) = 50(1 - e^{-t/20})$. $r(20) = 50(1 - e^{-1}) \approx 31.6\,\text{Hz}$ (63%). $r(60) = 50(1 - e^{-3}) \approx 47.5\,\text{Hz}$ (95%).

EXAMPLE · P5: DECOUPLED SOLUTION

$\dot{\vec{x}} = M\vec{x}$ with $M = \begin{pmatrix} -1 & 2 \\ 0 & -3 \end{pmatrix}$. Find the eigenvalues and eigenvectors, and decide on stability (a) from the eigenvalues and (b) from Tr/Det.

Solution. M is triangular, so the eigenvalues are its diagonal entries: $\lambda_1 = -1$, $\lambda_2 = -3$. $\lambda_1 = -1$: $(M + I)\vec{u} = 0 \Rightarrow \begin{pmatrix} 0 & 2 \\ 0 & -2 \end{pmatrix} \vec{u} = 0 \Rightarrow \vec{u}_1 = (1, 0)$. $\lambda_2 = -3$: $(M + 3I)\vec{u} = 0 \Rightarrow \begin{pmatrix} 2 & 2 \\ 0 & 0 \end{pmatrix} \vec{u} = 0 \Rightarrow \vec{u}_2 = (1, -1)$. So $\vec{x}(t) = C_1 e^{-t}(1, 0) + C_2 e^{-3t}(1, -1)$. (a) Both eigenvalues are negative $\Rightarrow$ converges to $\vec{0}$. (b) Tr $= -4 < 0$, Det $= 3 > 0 \Rightarrow$ converges. ✓

EXAMPLE · P6: TR/DET ONLY

For which values of w does $\dot{\vec{x}} = \begin{pmatrix} -1 & w \\ w & -1 \end{pmatrix} \vec{x}$ converge to $\vec{0}$?

Solution. Tr $= -2 < 0$ always. Det $= 1 - w^2 > 0 \iff |w| < 1$. So it converges exactly when $|w| < 1$. (Check with $w = 2$: the eigenvalues are -1 ± 2, i.e. 1 and -3; one is positive $\Rightarrow$ diverges.)

SECTION 28

Cheat sheet

WEEK 01 ON ONE PAGE

- **Neuron**: soma (cell body), axon (output), dendrites (input). Rest $\approx -70\,\text{mV}$, threshold $\approx -50\,\text{mV}$.
- **Spike**: $\sim$1 ms, stereotyped, all-or-none. Spikes are the currency of communication; their shape carries no information.
- **Chemical synapse**: spike $\to$ neurotransmitter $\to$ receptors $\to$ ion channels. Excitatory = depolarize, inhibitory = hyperpolarize.
- **Dale**: all outgoing synapses of a neuron have one sign (one column of J).
- **Rate model**: $r_i = g\left(\sum_j J_{ij} r_j + b_i\right)$, J_{ij} = from j to i; with dynamics $\tau \dot{r}_i = -r_i + g(\cdot)$.
- **Scales**: 1 nm (molecules) to 1 m (CNS); 1 ms (spike) to 10^9 s (long-term memory). Human: 10^{11} neurons, 10^{14}–10^{15} synapses.
- $\dot{n} = an \Rightarrow n(t) = n_0 e^{at}$; grows if $a > 0$, decays if $a < 0$; doubling/halving time $\ln 2/|a|$.
- $\dot{n} = an + I(t) \Rightarrow n(t) = n_0 e^{at} + \int_0^t I(\tilde{t}) e^{a(t-\tilde{t})} d\tilde{t}$ (a convolution $I * e^{at}$); constant I, $a < 0$: $n \to -I/a$.
- **Fixed point**: $\dot{x} = 0$. τ = time to reach 63% of the final value.
- $\dot{\vec{x}} = M\vec{x}$, $M = UDU^{-1}$ (U: right eigenvectors as columns; U^{-1}: left eigenvectors as rows, scaled so $\vec{w}_k^{\top} \vec{u}_k = 1$). $\Rightarrow \vec{x}(t) = \sum_k C_k e^{\lambda_k t} \vec{u}_k$.
- **Stable** $\iff$ all $\lambda_k < 0$. 2×2: stable $\iff$ Tr < 0 and Det > 0.